



Family 1: Decoder-backend Change decoder algorithm; hold prior and deadline fixed

	Decoder algorithm	Prior distribution	Deadline budget (μ s)	Examples
C ₀	Corr. matching	SI1000	No deadline	CM → Tesseract PM → CM
	≠	=	=	
C ₁	Tesseract	SI1000	No deadline	PM → Tesseract

Family 2: Decoder-prior Sweep prior distribution; hold decoder and deadline fixed

	Decoder algorithm	Prior distribution	Deadline budget (μ s)	
C ₀	Tesseract	dem_simple	No deadline	5,724 prior variants across 3 families
	=	≠	=	
C ₁	Tesseract	dem_corr / iso / sh	No deadline	19.08M paired shots

Family 3: Runtime-deadline Apply measured service-time budget; hold decoder and prior fixed

	Decoder algorithm	Prior distribution	Deadline budget (μ s)	
C ₀	Corr. matching	SI1000	∞ (no limit)	4 μ s → 93.6% miss
	=	=	≠	5 μ s → partial miss
C ₁	Corr. matching	SI1000	D ∈ {4,5,6,7,8} μ s	7 μ s → ≈ 0% miss